

In the central nervous system, myelin produced by oligodendrocytes (glial cells) functions as an insulating sheath surrounding certain nerve fibers. To identify candidate genes involved in the myelination process, researchers collected oligodendrocytes from zebrafish and purified the messenger RNA (mRNA) transcripts expressed by these glial cells. The mRNA was converted to complementary DNA (cDNA) by reverse transcriptase, and the cDNA was fluorescently labeled and assessed for hybridization on a microarray. The expression of mRNA in other zebrafish cells was similarly measured.

Transcripts detected in zebrafish oligodendrocytes at levels greater than 3 times those found in other cells were selected as candidate myelination genes. Some of the detected genes, including *plp1a*, *Sox10*, and *mbp*, were previously known to be specific to oligodendrocytes, validating the procedure. The cDNA of newly identified candidates was amplified by PCR, and the ends were digested with the restriction enzymes *EcoRI* and *XhoI*. The pSKII vector, shown in Figure 1, was then also digested with *EcoRI* and *XhoI*, and genes were ligated into the multiple cloning site (MCS). The positions of various restriction sites in the MCS are shown in Figure 2.

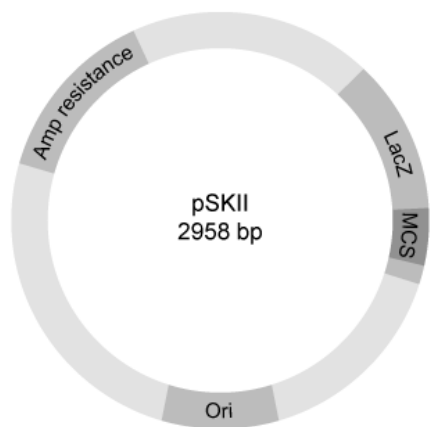


Figure 1 Overview of the pSKII vector (plasmid)

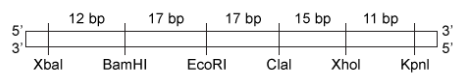


Figure 2 The pSKII MCS, showing the restriction sites of several restriction enzymes

Table 1 shows the functional parameters of several restriction enzymes.

Table 1

Restriction enzyme	Restriction site	Optimal enzyme conditions		
		pH	[NaCl] (mM)	[KAcO] (mM)
BamHI	GGATCC CCTAGG	7.9	100	0
ClaI	ATCGAT TAGCTA	7.9	0	50
EcoRI	GAATTC CTTAAG	7.9	0	50
KpnI	GGTACC CCATGG	7.0	0	0
XbaI	TCTAGA AGATCT	7.9	50	0
XhoI	CTCGAG GAGCTC	7.9	0	50

To confirm that the candidate genes originated in oligodendrocytes, complementary RNA (cRNA) strands were synthesized from the cloned plasmids and hybridized to fixed zebrafish sections. One of the genes that localized to oligodendrocytes, known as *cldnk*, produces a 915 base pair mRNA transcript from two exons. The *cldnk* gene is involved in tight junction formation and may be required for myelin sheath development around axons.

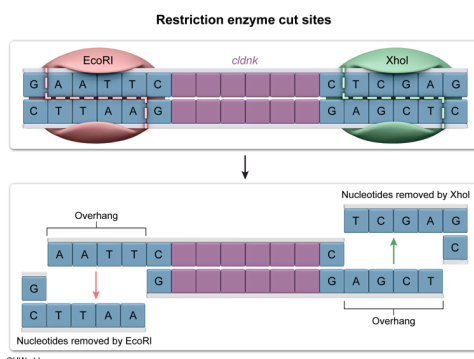
Which of the following diagrams shows the structure of the cloned *cldnk* gene after digestion with EcoRI and XhoI?

- ☐ A.
- GAATTC cldnk gene C
G GAGCTC
- [16%]
- ☒ B.
- AATTC cldnk gene C
G GAGCT
- [60%]
- ☐ C.
- AATTC cldnk gene T
G AGATC
- [17%]
- ☐ D.
- GAATTC cldnk gene T
G AGATCT
- [5%]

Correct

60% answered correctly

Explanation:



Restriction enzymes cleave DNA molecules at specific nucleotide sequences, known as **restriction sites**. Each restriction enzyme has a **unique restriction site** that typically consists of a DNA **palindrome**, in which the sequence of one strand is the reverse of its complement strand. If a restriction enzyme cuts anywhere other than directly in the middle of the palindrome, a few nucleotides will be left on the end of the DNA strand without any complementary bases paired with them. These leftover nucleotides are known as **overhangs** or **sticky ends** because they can anneal or "stick to" their complementary bases in another DNA strand.

The passage states that the *cldnk* gene was digested by EcoRI and XhoI. Table 1 shows that EcoRI cuts between the G and the A in GAATTC, leaving an AATT overhang, and XhoI cuts between the C and the T of

CTCGAG, leaving a TCGA overhang. Therefore, the digested gene should have an **AATT overhang** on one end and a **TCGA overhang** on the other, as shown in Choice B. (Note: Because the XhoI overhang is shown as the bottom strand, it is read from right to left [5' to 3'].)

(Choice A) EcoRI breaks the bond between the G and the A in GAATTC, so the G would not be present on the overhang after the digest occurs. Similarly, the terminal C would not be present in the overhang after XhoI digests the CTCGAG sequence.

(Choice C) Digestion of one end of the DNA by XbaI (*not* XhoI or EcoRI) would leave the CTAG overhang shown in this diagram (reading from right to left at the bottom of the diagram).

(Choice D) This diagram shows the wrong overhang for the EcoRI site (GAATT instead of AATT) and the wrong overhang for an XbaI site (TCTAG instead of CTAG).

Educational objective:

Restriction enzymes cleave DNA molecules at specific sites, known as restriction sites. Each enzyme has a unique restriction site and may cut the DNA such that an overhang, or sticky end, is left on the end of the strand.

Time spent: 0 sec

Subject:
Biology

QID: 402589

System:
Molecular Biology

In the central nervous system, myelin produced by oligodendrocytes (glial cells) functions as an insulating sheath surrounding certain nerve fibers. To identify candidate genes involved in the myelination process, researchers collected oligodendrocytes from zebrafish and purified the messenger RNA (mRNA) transcripts expressed by these glial cells. The mRNA was converted to complementary DNA (cDNA) by reverse transcriptase, and the cDNA was fluorescently labeled and assessed for hybridization on a microarray. The expression of mRNA in other zebrafish cells was similarly measured.

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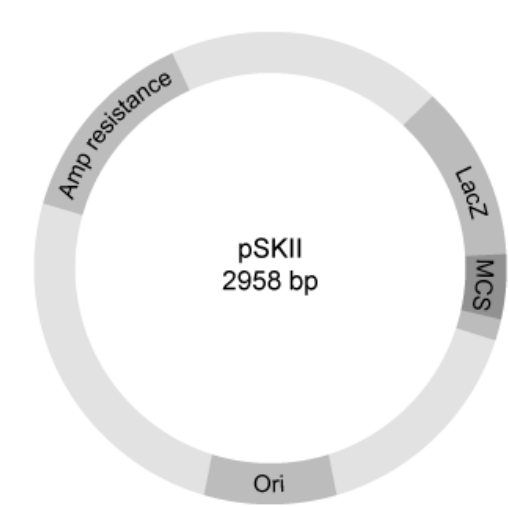


Figure 1 Overview of the pSKII vector (plasmid)

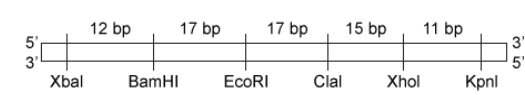


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		pH	[NaCl] (mM)	[KAcO] (mM)
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Clal	ATCGAT TAGCTA	7.9	0	50
EcoRI	GAATTC CTTAAG	7.9	0	50
KpnI	GGTACC CCATGG	7.0	0	0
XbaI	TCTAGA AGATCT	7.9	50	0
XhoI	CTCGAG GAGCTC	7.9	0	50

To confirm that the candidate genes originated in oligodendrocytes, complementary RNA (cRNA) strands were synthesized from the cloned plasmids and hybridized to fixed zebrafish sections. One of the genes that localized to oligodendrocytes, known as *cldnk*, produces a 915 base pair mRNA transcript from two exons. The *cldnk* gene is involved in tight junction formation and may be required for myelin sheath development around axons.

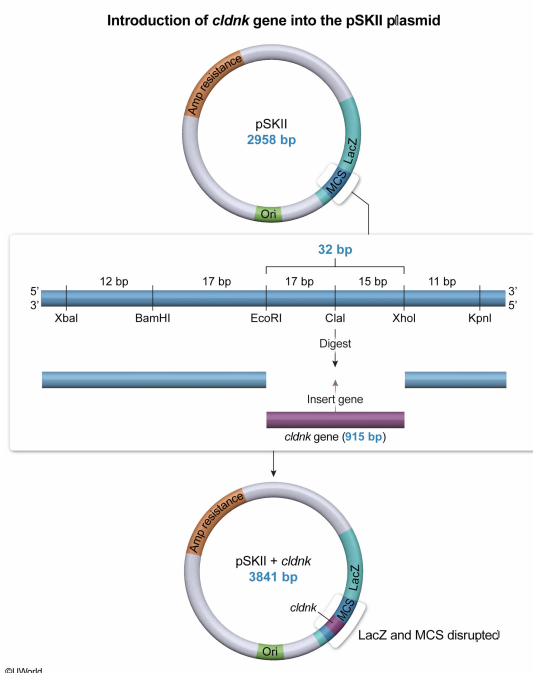
What is the size of the pSKII plasmid after it is digested by EcoRI and XhoI and the *cldnk* gene is inserted?

- ✓ ☒ A. 3,841 base pairs [51%]
☐ B. 3,856 base pairs [7%]
☐ C. 3,862 base pairs [10%]
☐ D. 3,873 base pairs [30%]

Correct

51% answered correctly

Explanation:



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Plasmids—small, **circular DNA molecules**—are commonly present in bacteria but are also found in some eukaryotic cells. Bacteria can exchange plasmids naturally through [conjugation](#). Artificial plasmids are commonly used in laboratories to amplify or express a gene of interest. A gene may be inserted into a plasmid through the [following steps](#):

1. Digest the plasmid with restriction enzymes that cut only the plasmid at the multiple

cloning site. If two restriction enzymes are used, any nucleotides between the two cut sites will be removed from the plasmid.

2. Digest the gene of interest with the same enzymes as the plasmid to ensure that the plasmid and the gene will **anneal** to each other. **PCR** may be necessary to **introduce cut sites** into the ends of the gene.
3. Incubate the digested plasmid and the digested gene together, along with **DNA ligase** to link them.

Figure 1 shows that prior to insertion of *cldnk*, the pSKII plasmid contained 2,958 base pairs. Figure 2 shows that the distance between the EcoRI and XhoI restriction sites is $17 + 15 = 32$ base pairs; these bases will be *removed* by the digest. The passage states that the *cldnk* gene produces a processed transcript that is 915 base pairs long, so insertion of *cldnk* cDNA increases the plasmid size by 915 base pairs. Therefore, the final size of the plasmid after *cldnk* is inserted can be calculated as $2,958 - 32 + 915 = 3,841$.

(Choice B) The plasmid would have **3,856 base pairs** if the ClaI site had been used instead of XhoI.

(Choice C) The plasmid would have **3,862 base pairs** if the KpnI site had been used instead of EcoRI.

(Choice D) The plasmid would have **3,873 base pairs** if the 32 bases between the EcoRI site and the XhoI site had not been removed. This would happen if only one restriction enzyme (eg, EcoRI) had been used.

Educational objective:

Plasmids—small, circular DNA molecules that can carry a small number of genes—are often used in laboratories to amplify and express genes of interest. A gene may be inserted into a plasmid by digesting the gene and the plasmid with the same restriction enzymes, followed by ligation.

Time spent:0 sec
Subject:
Biology

QID:402590
System:
Molecular Biology

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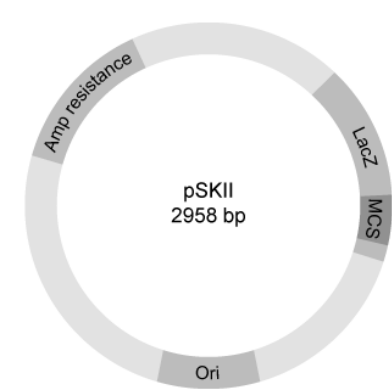


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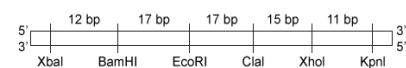


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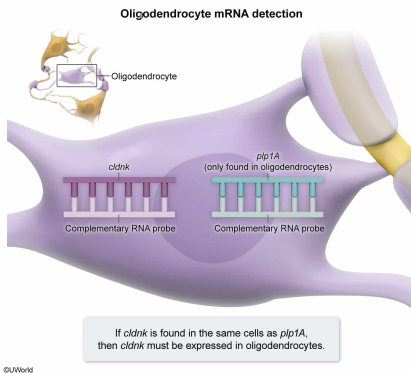
What observation could have led researchers to conclude that *cldnk* is most likely expressed in oligodendrocytes?

- ☐ A. A probe made of *cldnk* mRNA hybridized to RNA in cells that express *Sox10*. [13%]
- ✓ ☒ B. An RNA probe that was complementary to *cldnk* mRNA hybridized in cells that express *plp1a*. [43%]
- ☐ C. A *cldnk* cDNA probe obtained from oligodendrocytes bound to a microarray with higher affinity than the same probe obtained from other cells. [30%]
- ☐ D. A cDNA probe corresponding to *cldnk* hybridized to the nuclei of oligodendrocytes but not the cytosol. [12%]

Correct

42% answered correctly

Explanation:



In genetic studies, **hybridization** refers to the **annealing** of two complementary nucleic acid strands. Specifically, messenger RNA (mRNA) is a single-stranded nucleic acid that is present only in the cytosol of cells that **express** the corresponding gene. To determine whether a cell expresses a specific gene, tissues may be incubated with a labeled RNA strand that is **complementary** to the mRNA of interest. This labeled RNA hybridizes with the mRNA in cells that express it

but is washed away from other cells, allowing researchers to visualize which cells express a particular gene.

For a **cell that expresses a gene** of interest, one way to **determine the cell type** is to identify the presence of **another mRNA** that is known to be **specific to the cell type of interest**. The passage gives the names of three genes that are specific to oligodendrocytes: *plp1a*, *Sox10*, and *mbp*. If *cldnk* mRNA is found in the same cells as the mRNA of any of those genes, then *cldnk* must be expressed in oligodendrocytes. Therefore, if a **probe** that is **complementary to *cldnk* mRNA hybridizes to cells that also express *plp1A***, then *cldnk* is expressed in oligodendrocytes.

(Choice A) Although *Sox10* is specific to oligodendrocytes, the probe must be **complementary** to the mRNA of interest for hybridization to occur. If the probe is made of *cldnk* mRNA, it will be **identical**, not complementary, to *cldnk* mRNA in the cell and will not hybridize.

(Choice C) The affinity of *cldnk* complementary DNA for the microarray will be the **same**, regardless of which cells the cDNA came from. *Cldnk* cDNA derived from oligodendrocytes binds to a **greater extent** than the same cDNA derived from other cells because more of it is present, **not** because it has higher affinity.

(Choice D) Gene expression is mediated by mRNA, which is exported to the cytosol for translation. If the probe binds to the nucleus but not the cytosol, it would indicate the presence of RNA that is **not** being sent to the cytosol for expression.

Educational objective:

In genetic studies, hybridization is the annealing of two complementary nucleic acids. The presence of a particular messenger RNA within a specific cell type can be assessed by hybridization to a complementary probe.

Time spent: 0 sec
Subject:
Biology

QID: 402591
System:
Molecular Biology

In the central nervous system, myelin produced by oligodendrocytes (glial cells) functions as an insulating sheath surrounding certain nerve fibers. To identify candidate genes involved in the myelination process, researchers collected oligodendrocytes from zebrafish and purified the messenger RNA (mRNA) transcripts expressed by these glial cells. The mRNA was converted to complementary DNA (cDNA) by reverse transcriptase, and the cDNA was fluorescently labeled and assessed for hybridization on a microarray. The expression of mRNA in other zebrafish cells was similarly measured.

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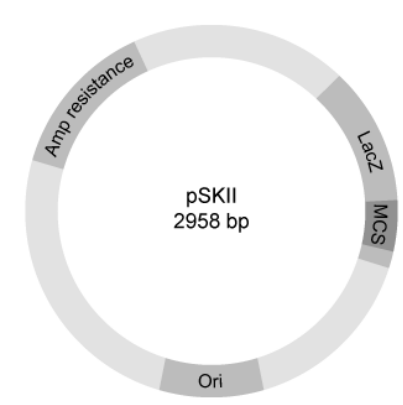


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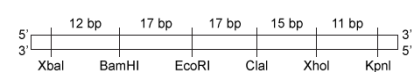


Figure 2 The pSKII MCS, showing the restriction sites of several restriction enzymes

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ClaI	A↓TCGAT TAGC↑TA	7.9	0	50
EcoRI	G↓AATTC CTTAA↑G	7.9	0	50
KpnI	GGTAC↓C C↓CATGG	7.0	0	0
XbaI	T↓CTAGA AGATC↑T	7.9	50	0
XhoI	C↓TCGAG GAGCT↑C	7.9	0	50

To confirm that the candidate genes originated in oligodendrocytes, complementary RNA (cRNA) strands were synthesized from the cloned plasmids and hybridized to fixed zebrafish sections. One of the genes that localized to oligodendrocytes, known as *cldnk*, produces a 915 base pair mRNA transcript from two exons. The *cldnk* gene is involved in tight junction formation and may be required for myelin sheath development around axons.

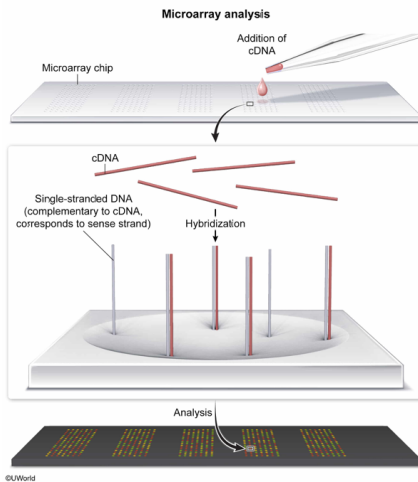
Microarrays are chips that contain hundreds of microscopic wells, each of which can detect a distinct nucleic acid. Prior to exposure to cDNA, the wells of the microarray described in the passage most likely contained:

- ☐ A. double-stranded fragments of the zebrafish genome. [10%]
- ☐ B. proteins isolated from zebrafish samples. [6%]
- ☐ C. antibodies against the cDNA generated by the reverse transcriptase enzyme. [19%]
- ☒ D. single-stranded DNA with the same sequences as the sense strands of zebrafish gene exons. [62%]

Correct

63% answered correctly

Explanation:



A gene within genomic DNA consists of **two strands**: a **sense strand** (coding strand) and an **antisense** (noncoding)

strand. RNA is produced when RNA polymerase uses the antisense strand as a template to make a new nucleic acid. The newly synthesized **RNA strand** accordingly has the **same sequence as the sense strand** of the DNA (with the exception that RNA contains uracil instead of thymine).

During the **formation of cDNA**, the enzyme **reverse transcriptase** produces DNA using **RNA as a template**. The **cDNA** strand made during this process is complementary to the RNA from which it was made, so it is **complementary to the sense strand** of the genomic DNA (ie, the cDNA is an antisense strand).

The cDNA in the passage was obtained from zebrafish; therefore, to detect the cDNA, the microarray must contain **single-stranded DNA** from zebrafish genes. Because cDNA is complementary to the sense strand of genomic DNA, the single-stranded DNA on the array must have the same sequence as the **sense strands of zebra fish genes**, allowing them to hybridize with cDNA.

(Choice A) Double-stranded DNA is **already hybridized** and would be unable to anneal to the cDNA probe. The cDNA would be washed away from all wells, and no signal would be detected.

(Choice B) Proteins perform many different functions within cells, but only a subset (eg, **transcription factors**) bind to DNA. Accordingly, the cDNA would be **unlikely** to hybridize with zebrafish proteins, and proteins would **not** be useful in analyzing gene expression via a microarray.

(Choice C) Antibodies typically bind proteins, **not** DNA. Antibodies may be used in protein identification techniques such as **ELISA** but would **not** hybridize with cDNA.

Educational objective:

During complementary DNA (cDNA) synthesis, reverse transcriptase uses messenger RNA (mRNA) as a template to synthesize a new single-stranded DNA molecule. Because mRNA has the same sequence as the **sense** strand of genomic DNA (using uracil instead of thymine), cDNA will have the same sequence as the **antisense** strand. Accordingly, cDNA hybridizes with single-stranded DNA that has the same sequence as the sense strand of genomic DNA.

Time spent: 0 sec
Subject:
Biology

QID: 402592
System:
Molecular Biology

In the central nervous system, myelin produced by oligodendrocytes (glial cells) functions as an insulating sheath surrounding certain nerve fibers. To identify candidate genes involved in the myelination process, researchers collected oligodendrocytes from zebrafish and purified the messenger RNA (mRNA) transcripts expressed by these glial cells. The mRNA was converted to complementary DNA (cDNA) by reverse transcriptase, and the cDNA was fluorescently labeled and assessed for hybridization on a microarray. The expression of mRNA in other zebrafish cells was similarly measured.

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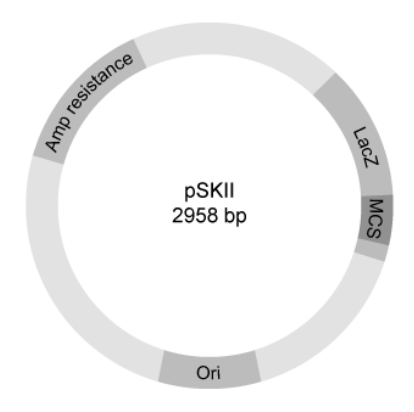


Figure 1 Overview of the pSKII vector (plasmid)

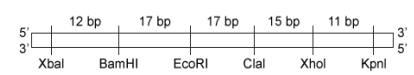


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To confirm that the candidate genes originated in oligodendrocytes, complementary RNA (cRNA) strands were synthesized from the cloned plasmids and hybridized to fixed zebrafish sections. One of the genes that localized to oligodendrocytes, known as *cldnk*, produces a 915 base pair mRNA transcript from two exons. The *cldnk* gene is involved in tight junction formation and may be required for myelin sheath development around axons.

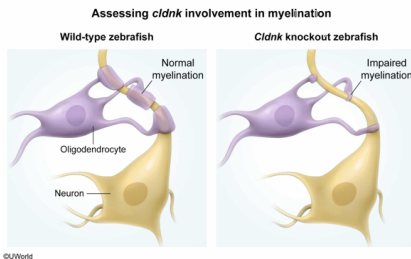
Which experiment could confirm that the *cldnk* gene product is required for myelination?

- ☐ A. Compare myelin formation in wild-type zebrafish to formation in zebrafish that overexpress cloned *cldnk* [3%]
- ☐ B. Overexpress *cldnk* in zebrafish cells other than oligodendrocytes and observe whether those cells produce myelin [3%]
- ✓ ☒ C. Compare myelin formation in wild-type zebrafish to that in zebrafish with *cldnk* knocked out [90%]
- ☐ D. Overexpress a form of *cldnk* that inhibits the wild-type gene in cells other than oligodendrocytes and observe whether myelin forms [2%]

Correct

90% answered correctly

Explanation:



Knockout organisms have **one or more genes** that have been made **inactive**. Genes may be knocked out by introducing mutations that result in a truncated gene product, a frameshifted gene product, or no gene product, each of which results in genes that produce nonfunctional proteins. The **biological function** of a gene can be inferred by knocking it out and observing the differences between wild-type organisms and knockout organisms.

If *cldnk* is required for wild-type myelination of oligodendrocytes, then knocking the gene out should result in impaired or abnormal myelination. Therefore, **comparing myelination** of oligodendrocytes in **wild-type zebrafish** to myelination in ***cldnk* knockout zebrafish** would show whether *cldnk* is required for proper myelin formation.

(Choice A) If *cldnk* is involved in myelination, it is *possible* that overexpression of *cldnk* would cause increased myelination. However, it is also possible that endogenous *cldnk* already yields maximum activity, in which case excess *cldnk* would have no effect. Therefore, overexpression of *cldnk* would not necessarily give any information about its role in myelination.

(Choice B) Although *cldnk* expression in cells other than oligodendrocytes could cause myelin to form in those cells, it is unlikely because other gene products are most likely required for myelination as well.

(Choice D) A form of the *cldnk* gene that inhibits activity of the wild-type gene (ie, a dominant negative gene) could show whether *cldnk* is required for myelination if it were expressed in *oligodendrocytes* but not in cells *other than* oligodendrocytes.

Educational objective:

A gene's biological function can be inferred by comparing the differences in organisms with the gene knocked out (inactivated) to wild-type organisms.

Time spent: 0 sec
Subject:
Biology

QID: 402593
System:
Molecular Biology

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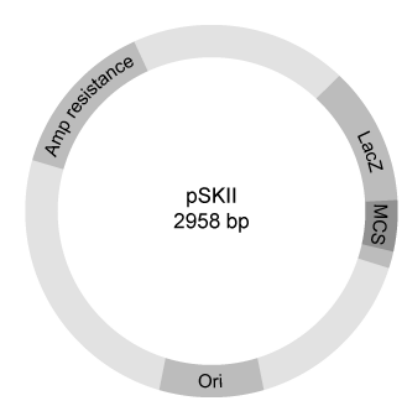


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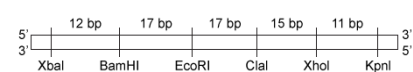


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XbaI	T CTAGA AGATCT A	7.9	50	0
XhoI	C TCGAG GAGCTC A	7.9	0	50

To confirm that the candidate genes originated in oligodendrocytes, complementary RNA (cRNA) strands were synthesized from the cloned plasmids and hybridized to fixed zebrafish sections. One of the genes that localized to oligodendrocytes, known as *cldnk*, produces a 915 base pair mRNA transcript from two exons. The *cldnk* gene is involved in tight junction formation and may be required for myelin sheath development around axons.

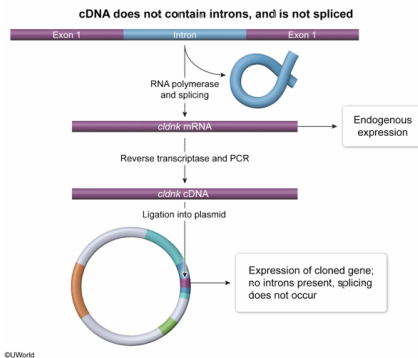
How would expression of cloned *cldnk* differ from expression of the endogenous *cldnk* gene?

- ✓ ☒ A. Expression of the cloned gene does not involve splicing whereas expression of the endogenous gene does [66%]
- ☐ B. The cloned gene is expressed at lower levels than the endogenous gene [9%]
- ☐ C. The cloned gene produces a protein with a different sequence than the endogenous gene [5%]
- ☐ D. Expression of the cloned gene produces a mature mRNA molecule with a different coding region than expression of the endogenous gene [18%]

Correct

66% answered correctly

Explanation:



Genes within a eukaryotic chromosome typically consist of **exons and introns**, both of which are transcribed by RNA polymerase to form **pre-mRNA**. After transcription, pre-mRNA matures through [several processes](#), including 5' capping by 7-methylguanosine, 3' polyadenylation, and [splicing](#), to **remove introns**.

Complementary DNA (cDNA) is derived from mature mRNA through **reverse transcription**. Because the mature mRNA has already been spliced (ie, introns were removed), the cDNA will *not* contain any introns. The cDNA can be cloned into expression vectors such as plasmids, which can be introduced into embryonic cells and integrated into the genome. These cloned genes can then be expressed in the organism. However, because the cloned gene does not contain any introns, its **expression will not include splicing**.

(Choice B) Cloned genes may be expressed at lower levels, the same levels, or higher levels than endogenous genes, depending on several factors including how the cloning was done, how expression was induced, and the type of cell being observed.

(Choices C and D) The cDNA was derived from endogenous mature mRNA, so it will yield mature mRNA with a coding region that is *identical* to that of the endogenous gene. Because mRNA determines the sequence of the protein to be produced, the cloned gene yields the *same* protein as the endogenous mRNA from which it was derived.

Educational objective:

RNA polymerase transcribes both exons and introns to form pre-mRNA. During RNA processing, introns are removed by splicing to yield mature mRNA. Complementary DNA is generated from mature mRNA and does not contain introns, so it is not spliced during expression.

Time spent: 0 sec
Subject:
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System:
Molecular Biology